



Ahsanullah University of Science & Technology

Department of Computer Science & Engineering

Course No : CSE 2214
Course Title : Assembly Language Programming Sessional
Assignment No : 06

Date of Submission : Thursday, 1st October, 2026

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Submitted By-

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Section : A

Write an assembly language program that displays a sequence formed by interleaving the last two digits of your studentID with the last two letters of your name(digit-letter-digit-letter-...), one by one, on consecutive dot matrix displays. After the sequence has been shown, the program should continuously scroll the pattern from right to left until the entire sequence reaches the leftmost position, then reverse the direction and scroll it from left to right. Repeat the process continuously with suitable delays.

For example, If the Student's name is Ahmed Abdullah, and his ID is 00724105101027, the sequence to be displayed should be 2A7H

```
;00724105101008
.STACK 100H
.MODEL SMALL
.DATA
FONT1 DB 0111110B,1000001B,1000001B,1000001B,0111110B
      DB 1111100B,0001010B,0001001B,0001010B,1111100B
      DB 0111110B,1001001B,1001001B,1001001B,0111110B
      DB 1111111B,0000100B,0001000B,0010000B,1111111B
.CODE

MAIN PROC
    MOV AX,@DATA
    MOV DS,AX

    MOV DX,2000H
    MOV CX,20

    MOV BX,2000H
RIGHT:
    MOV SI, OFFSET FONT1
    MOV CX,20
    MOV DX,BX
L1:

    MOV AL,[SI]
    OUT DX,AL
    INC SI
    INC DX
    DEC CX
    JNZ L1
    CALL DELAY
    CALL CLEAR
    CMP BX,2014H

    JE LEFT
    INC BX
    JMP RIGHT
LEFT:
    MOV DX,BX
```

```
ADD DX,19
MOV SI,OFFSET FONT1+19
MOV CX,20
L2:
MOV AL,[SI]
OUT DX,AL
```

```
DEC DX
DEC SI
DEC CX
JNZ L2
CALL DELAY
CALL CLEAR
```

```
CMP BX,2000H
JE RIGHT
DEC BX
JMP LEFT
MAIN ENDP
```

```
DELAY PROC
MOV AX,15
L3:
DEC AX
CMP AX,0
JNZ L3
RET
DELAY ENDP
```

```
CLEAR PROC
MOV DX,2000H
MOV AX,0
MOV CX,40
LO:
OUT DX,AL
INC DX
LOOP LO
RET
CLEAR ENDP
END MAIN
```